

**YEAR 9 SCIENCE
CHEMICAL SCIENCES
REVISION - TEST 2**

1. (a) List the properties of an **acid**.

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- (b) List the properties of a **base**.

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- (c) What is a **salt**?

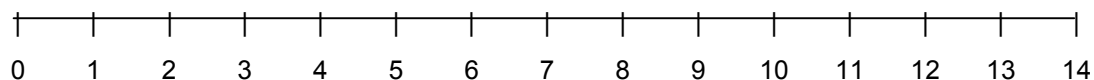
- (d) Classify the following as **ACID**, **BASE** or **SALT**. Use the table below.

ZnCl₂ HCl copper hydroxide H₂SO₄ sodium sulphate calcium acetate
CH₃COOH MgO KOH CuSO₄ HNO₃ aluminium nitrate

ACID	BASE	SALT

2. Label the following diagram of the pH scale with the following.

Colour of universal indicator (write in the colours), weak acid, weak base, neutral, strong acid, strong base.



3. Complete the following table, giving **estimates on the pH where necessary**, the **colour of the universal indicator** and whether the substance is a **strong/weak acid**, a **strong/weak base** or **neutral**.

SUBSTANCE	pH	UNIVERSAL INDICATOR (colour)	ACID / BASE/ NEUTRAL (strong / weak)
urine		yellow	
pool water	8		
milk			neutral
lemonade		light red	
drain cleaner	14		
vinegar			weak acid

4. (a) Complete the following table for the acids and the salts they produce.

ACID	FORMULA	SALT PRODUCED
hydrochloric		
nitric		
sulphuric		

- (b) **Name the salt** produced when the following substances are added to an acid.

REACTION	NAME OF SALT PRODUCED
aluminium hydroxide + sulphuric acid	
copper + nitric acid	
calcium hydrogencarbonate + hydrochloric acid	
zinc carbonate + acetic acid	
iron + nitric acid	
magnesium oxide + hydrochloric acid	

5. Complete the following general word equations for acid reactions.

ACID + METAL $\rightarrow$ _____ + _____

ACID + BASE $\rightarrow$ _____ + _____

ACID + CARBONATE $\rightarrow$ _____ + _____
+ _____

6. Complete the following word equations for acid reactions

(a) magnesium + sulphuric acid $\rightarrow$ _____ + _____

(b) zinc oxide + nitric acid $\rightarrow$ _____ + _____

(c) sodium carbonate + hydrochloric acid $\rightarrow$ _____ + _____
+ _____

(d) copper oxide + sulphuric acid $\rightarrow$ _____ + _____

(e) zinc + hydrochloric acid $\rightarrow$ _____ + _____

(f) aluminium carbonate + sulphuric acid $\rightarrow$ _____ + _____
+ _____

7. **EXTENSION:** Balance the following equations.

(a) $\text{CuO}_{(\text{s})} + \text{HCl}_{(\text{aq})} \rightarrow \text{CuCl}_{2(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$

(b) $\text{Al}_{(\text{s})} + \text{F}_{2(\text{g})} \rightarrow \text{AlF}_{3(\text{s})}$

(c) $\text{Mg}(\text{NO}_3)_{2(\text{aq})} + \text{K}_3\text{PO}_{4(\text{aq})} \rightarrow \text{Mg}_3(\text{PO}_4)_{2(\text{s})} + \text{KNO}_{3(\text{aq})}$

(d) $\text{Fe}_2\text{O}_{3(\text{s})} + \text{CO}_{(\text{g})} \rightarrow \text{Fe}_{(\text{s})} + \text{CO}_{2(\text{g})}$